Project 2 - The Librarian

Requirements assessment and completion report

Assessment of the Tunisian Commerce RAG implementation against the CS496 AI Engineering project brief. Generated 21 October 2026.

Executive verdict. The project is a credible, tested RAG implementation with strong corpus, retrieval, citation, and evaluation work. It is not yet fully complete against the brief because public deployment, real-user feedback, and a qualifying 50% Week 8 optimization are still outstanding.

What is done well

Requirement area	Evidence and assessment
Real corpus	193-page 2024 IORT Tunisian Code de commerce; source provenance and SHA-256 verification are documented.
Corpus scale and QA set	109 approved questions: 89 answerable, 20 unanswerable, including multi-article questions with source evidence.
Indexing	Article-aware records preserve article numbers, legal hierarchy, page metadata, and stable IDs. BM25 and multilingual vector indexing are implemented.
Retrieval	BM25, vector, hybrid reciprocal-rank fusion, and deterministic reranking are implemented and compared.
Citations	Claims cite returned chunks; quotes are checked against supplied text; article, page, and source metadata are attached server-side.
Evaluation	Recall@k, any-gold hits, strict multi-evidence success, latency, citation metrics, abstention metrics, token usage, and cost formulas are present.
Engineering quality	The full suite passes: 204 tests passed, 26 skipped, 0 failures.
Documentation	README, design, optimization report, postmortem, deployment files, and contribution roles are documented.

Measured retrieval results

Method	Recall@10	p50 search	p95 search
BM25	83.15%	1.82 ms	2.82 ms
Vector	68.54%	20.56 ms	27.48 ms
Hybrid	87.08%	21.69 ms	27.63 ms
Hybrid + reranker	87.08%	24.17 ms	32.55 ms

Hybrid improves recall over BM25 at k=10, but costs more latency. The reranker did not improve k=10 coverage in the fixed evaluation.

What remains incomplete or weak

Gap	Current status and impact
Week 8 optimization target	Not demonstrated. No before/after experiment proves a 50% cost or p95 reduction with no more than two percentage points of accuracy loss.
Public deployment	The FastAPI service and Dockerfile exist, but there is no public URL. Docker image build was not completed because Docker Desktop's Linux engine was unavailable.
Real-user feedback	No target-user sessions or anonymous task outcomes were collected.

Gap	Current status and impact
Abstention quality	Generation evaluation found 13/20 correct unanswerable abstentions, 7 unsupported answers, and 31/89 false abstentions.
Faithfulness measurement	Quote occurrence and citation-ID matching are structural proxies, not semantic entailment or legal correctness.
Cost model	Token-cost formulas exist, but local electricity, hardware amortization, and operating costs were not measured.

Recommended final actions

1. Deploy the Dockerized service and record a public URL, model configuration, and health check.

2. Collect anonymous feedback from real target users using the documented five-point task questionnaire.

3. Choose one optimization, rerun the unchanged 109-question set before and after, and report latency, cost, answer quality, citations, and abstention.

4. Calibrate MIN_RETRIEVAL_SCORE on a development split, not on the final test set, and rerun generation evaluation.

5. Add structured human faithfulness labels or a documented judge protocol.

Referenced repository files: README.md, design.md, optimisation.md, postmortem.md, docs/PERSON_4_EVALUATION.md, eval/retrieval.py, eval/generation.py, src/api/app.py, and src/generation/service.py.